

Debug IBM i from VS Code



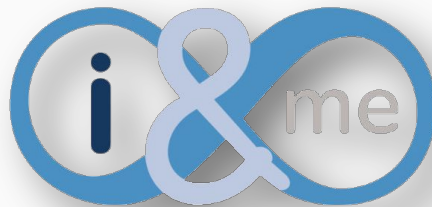
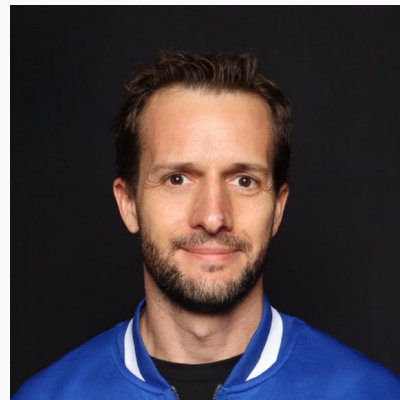
Ain't afraid of no bugs



Code for IBM i

Who's this guy?

- Sébastien “Seb” Julliand
- Code for IBM i *core team* member
- IBM i Pipeline Steps Jenkins plugin core maintainer
- IBM Champion since 2024
- R&D Tech Lead at i and me - Training and consulting



Code for IBM i

Thank you Seiden Group! 🙏

- For lending your LPAR for this session
- For applying the latest PTFs
- For Code for IBM i Fridays

CODE FOR IBM i FRIDAYS

Seiden Group hosts regular community gatherings for VS Code for i users to ask questions, share tips, and discuss open issues and enhancements.



Code for IBM i



Debugger Architecture 🤔

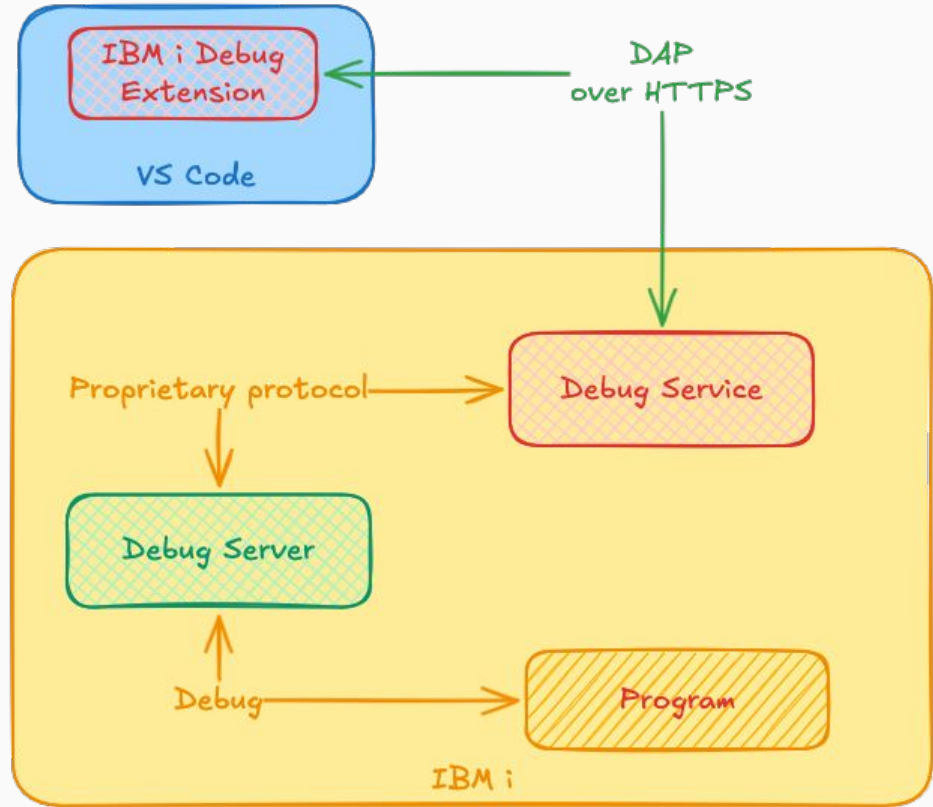
What's in there and how it works



Code for IBM i

The different parts

- VS Code extension
- Debug Service
- Debug Server



The Debug Server

- First used by CODE/400, then WDSC/RDI/RDP/RDi
- Start with STRDBGSVR
- Can be started from VS Code or iNav



The Debug Service

- A little history
 - V1 Introduced with...Merlin...
 - V2 added Service Entry Points
 - V3 uses DCM for certificate management
- Java product
 - RDi debugger wrapped in a web server
 - Installed under /QIBM/ProdData/IBMiDebugService
- Can be started from VS Code (for the moment) or iNav

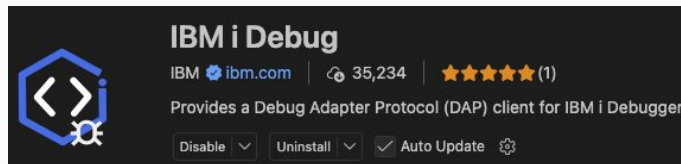


The IBM i Debug Extension

- Contributes IBM i debug support to VS Code
- Provides the interface to the Debug Service
 - Commands
 - IBM i debug configurations
- Code for IBM i calls its commands



Code for IBM i



Debug Service files

- /QIBM/ProdData/IBMiDebugService
 - Configuration (ports...)
 - Runtime
 - Updated by PTF
- /QIBM/UserData/IBMiDebugService
 - Client certificate
 - Runtime workspace
 - Log files
 - Keystore (server certificate)

```

v /QIBM/ProdData/IBMiDebugService
v bin
  ⚙ DebugService.env
  $ encryptKeystorePassword.sh
  $ startDebugService.sh
  $ stopDebugService.sh
  > configuration
  > plugins
  > plugins.update
  {} package.json
v /QIBM/UserData/IBMiDebugService
v certs
  🔒 debug_service.crt
  > configuration
  > startDebugService_workspace
  ≡ DebugService_log.txt
  ≡ TrustStore

```



It's all owned by IBM 🙄

- IBM i Debug extension is closed-source
- IBM i Debug Service is closed source
 - Not entirely owned by IBM i team
- Community space to report bugs and issues
 - <https://github.com/codefori/vscode-ibmi-debug-issues/issues>



Setting it up

That's the fun part



Code for IBM i

Keep it up to date

- <https://codefori.github.io/docs/developing/debug/configure/#required-ptfs>
- IBM releases fixes often
- Most issues fixed in 3.0.3



Code for IBM i

Required PTFs

To make use of the Debug Service, you need the following PTFs:

Version 3.0.3	Version 3.0.2	Version 3.0.1	Version
---------------	---------------	---------------	---------

- IBM i 7.6
 - Debug Service PTF SJ08472
 - Navigator for i PTF SJ08451
 - HTTP Group PTF SF99962 level 7 or higher
 - Java 17 LPP 5770JV1 Option 20 (not Java from Yum)

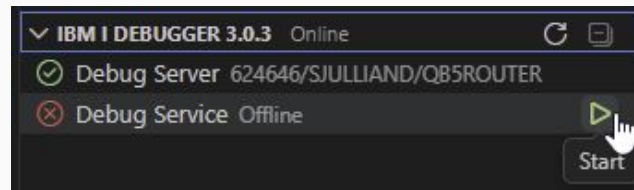
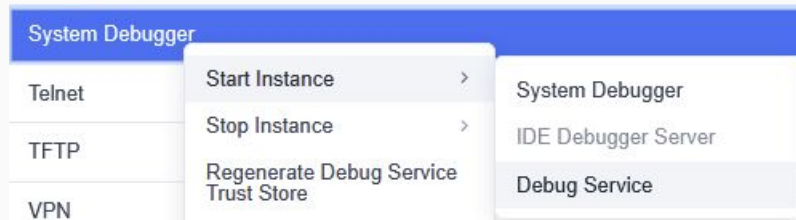
Generate the Keystore

- Go to iNav TCP/IP Servers
- Right-click System Debugger
- Select *Regenerate Debug Service Trust Store*
 - Requires access to DCM *SYSTEM keystore
 - Requires a valid certificate



Start the Debug Service

- Requires *ALLOBJ
- From iNav
 - Go to TCP/IP Servers
 - Right-click System Debugger
 - Select *Start Instance* -> *Debug Service*
- From VS Code
 - Go to *Run & Debug* view container
 - Expand *IBM i Debugger* view
 - Click on start



Debugging from VS Code

Finally !



Code for IBM i

What can be debugged

- ILE
 - Programs
 - Service Programs
- In batch mode (call & debug)
- Using Service Entry Points (trigger debug when called)

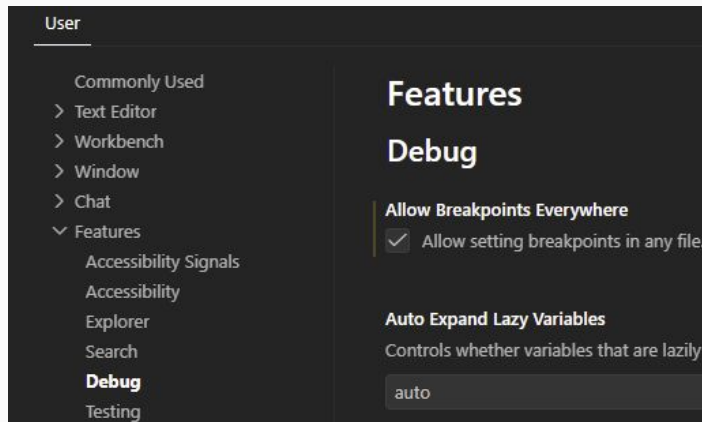


Check your settings

- Enable “Allow Breakpoints Everywhere”
 - Required to set breakpoints with DBGVIEW(*LIST)

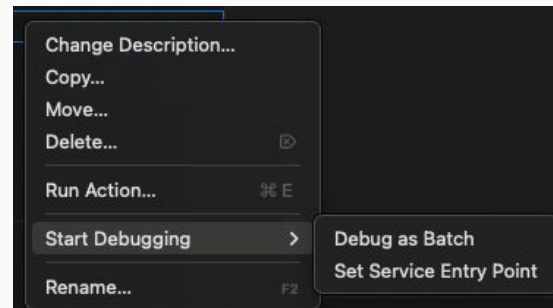
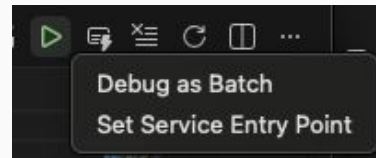


Code for IBM i



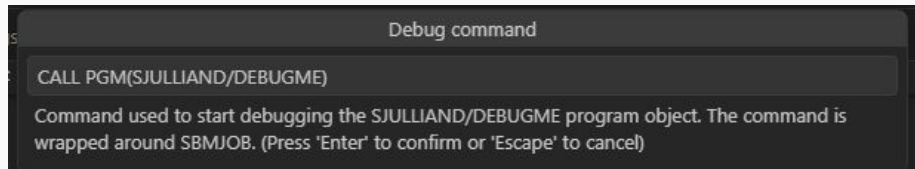
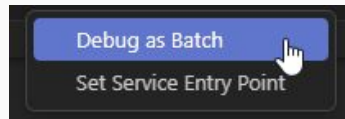
How to start a debug session

- From an opened editor
 - Play button in the title area
 - Supported language types: RPG, RPGLE, CL, COBOL, C
- From the Objects Browser
 - Define a filter to list *PGM and *SRVPGM
 - Right-click on target -> Start Debugging



Debug As Batch

- From the Object Browser
- From the Active Editor



IBM i debug configuration

- Defined in local `.vscode/launch.json` file
- Good for
 - Local projects
 - Repeated calls
- `launch.json` supports content assist
- Runs even if Code for i not connected

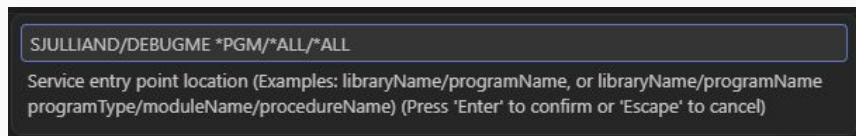
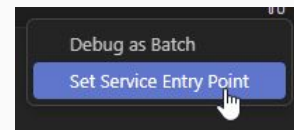
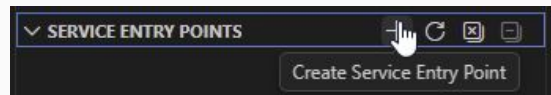


Code for IBM i

```
{  
  // Use IntelliSense to learn about possible attributes.  
  // Hover to view descriptions of existing attributes.  
  // For more information, visit: https://go.microsoft.com/fwlink/?linkid=830387  
  "version": "0.2.0",  
  "configurations": [  
    {  
      "type": "IBMiDebug",  
      "request": "launch",  
      "subType": "batch",  
      "name": "Debug DEBUGME",  
      "user": "SJULLIAND",  
      "password": "${command:AskForPassword}",  
      "host": "seiden.iinthecloud.com",  
      "port": 8005,  
      "library": "SJULLIAND",  
      "program": "DEBUGME",  
      "startBatchJobCommand": "SBMJOB CMD(CALL PGM(SJULLIAND/DEBUGME) PARM('HELLO'))",  
      "updateProductionFiles": false  
    }  
  ]  
}
```

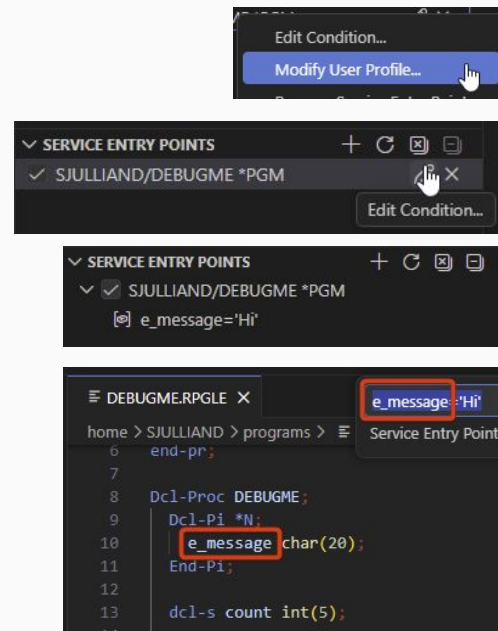
Service Entry Points - Create

- From the Service Entry Points view
- From the Object Browser
- From the Active Editor



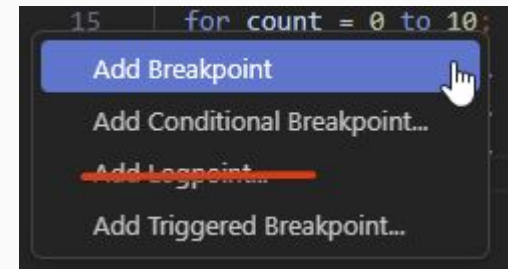
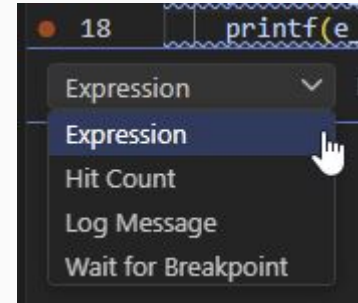
Service Entry Points - Edit

- Change user
 - Will start debug for a specific user only
- Add a condition
 - Activates the SEP when the condition is met
 - Condition must be on entry parameters



Setting breakpoints

- Normal breakpoints
 - Stops every time
- Conditional breakpoints/Expression
 - Stops when condition is met
- Triggered Breakpoint/Wait for breakpoint
 - Stops after another breakpoint has been
- Not supported
 - Logpoint/Log Message
 - Hit Count (wrong implementation)



Debug Panel

- Local variables for selected stack
- Watched variables
- Call stack

The screenshot displays the IBM i Debug Panel with the following sections:

- VARIABLES**
 - Locals**
 - `_QRNL_PSTR_E_MESSAGE` = SPP:EDEDE554B7001671
 - `COUNT` = 0
 - `E_MESSAGE` = HELLO
- WATCH**
 - `count` = 0
 - `e_message` = HELLO
- CALL STACK** (Pause on Step Over)
 - DEBUGME : DEBUGME : DEBUGME DEBUGME.LISTING 17:1
 - _QRNP_PEP_DEBUGME** : DEBUGME : DEBUGME DEBUGME.... 9:1
 - QCAPCMD : QCAPCMD : QCAPCMD QCAPCMD.STATEMENT 1:1
 - ProcessCommand : QRSEEXEC : EVFCEXEC EVFCEXEC.S... 1:1
 - main : QRSEEXEC : EVFCEXEC EVFCEXEC.STATEMENT 1:1
 - _C_pep : QRSEEXEC : EVFCEXEC EVFCEXEC.STATEME... 1:1
- BREAKPOINTS**
 - ☒ DEBUGME.LISTING 18



Any questions?

Don't be shy!



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Thank you !

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Meet me for Coffee